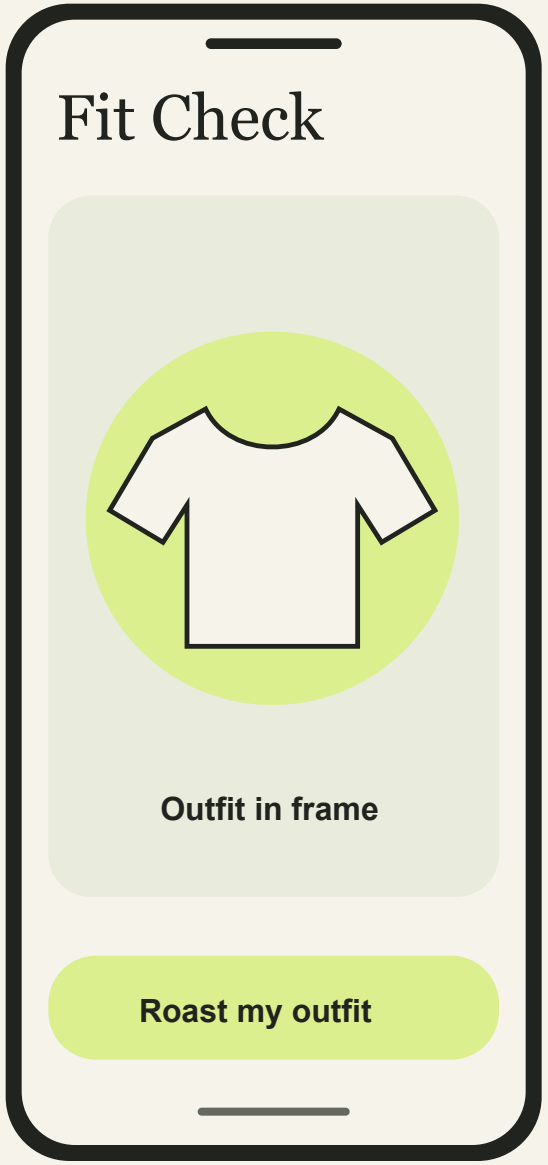
Your outfit.
A roast.
A better idea.

Point the camera. Ask for a roast.
Then ask how to make the outfit better.

"Hey Gemma, roast my outfit."

Gemma 4 E2B + LiteRT-LM. Everything on the phone.



Illustrative app view

The mirror talks back.

ACTUAL GEMMA OUTPUT / SAVED OUTFIT REPLAY

This ensemble screams 'I just woke up like this' with a palette that suggests a failed attempt at coordinated effort.

The blue and orange jacket looks like it was designed by a committee that couldn't agree on a primary color.

That white shirt is clearly trying too hard to be the neutral anchor in this aggressively colored disaster.

Then: "How would you make it better?"

Styling advice uses the same original outfit photo.

Selected, unedited model output. Screenshot and response are separate captures of the same outfit.

WHY BEST USE OF LITERT

A full multimodal experience

Gemma vision, audio and language on Pixel.

Deliberate runtime engineering

GPU generation; constrained JSON output.

Measured on the actual phone

Prepared contexts + 140 visual tokens.

Local by construction

No Internet permission; offline speech.

TWO SAVED-PHOTO INFERENCE RUNS

12.6 / 21.8 s baseline → 7.3 / 13.8 s tuned.
Small sample; excludes capture and speech.
Live voice and clothing accuracy need tuning.

TRACK 01 / MOBILE AI

One outfit. Two conversations.

A camera-first Gemma app that turns an outfit into spoken comedy, then useful advice.

01

Frame the outfit

Use the front or rear camera.
Talk or tap to start.

→

02

Hear the roast

An overall verdict, then jokes about visible clothing.

→

03

Ask for a better fit

Styling advice reuses the original outfit photo.

One original photo stays in the local session

WHY IT MERITS CONSIDERATION

Gemma connects vision, speech input and language in a playful mobile flow. The next question has context, so the experience continues beyond a single caption.

ON-DEVICE EVIDENCE

Camera → roast → speech → advice verified.
Buttons remain the reliable demo trigger;
live voice and clothing accuracy need tuning.

SPECIAL AWARD / BEST USE OF LITERT

The runtime is part of the product.

LiteRT-LM 0.18.1 runs Gemma vision, audio transcription and language generation on the Pixel.

PHOTO + AUDIO

CameraX / AudioRecord
Foreground capture

→

GEMMA 4 E2B / LITERT-LM

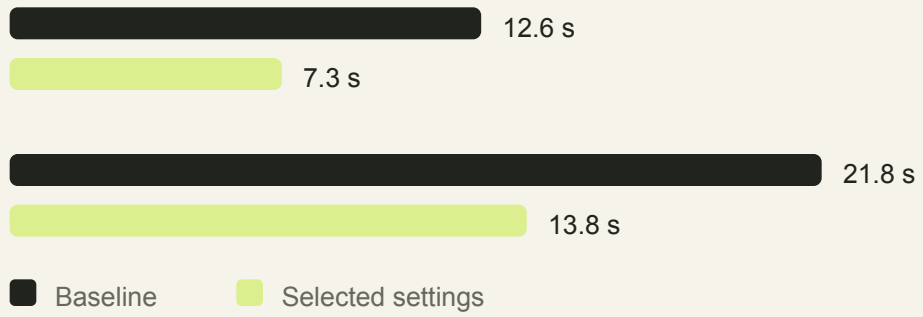
GPU language generation · CPU image/audio encoders
140 visual tokens · prepared system contexts
JSON-schema constrained response decoding

→

VALIDATE + SPEAK

Three lines / styling advice
Offline Android TTS

PROFILED ON THE ACTUAL PHONE



WHY IT MERITS CONSIDERATION

We measured setup and generation, compared runtime choices, and kept the settings that worked on our device. Source and tradeoffs are public.

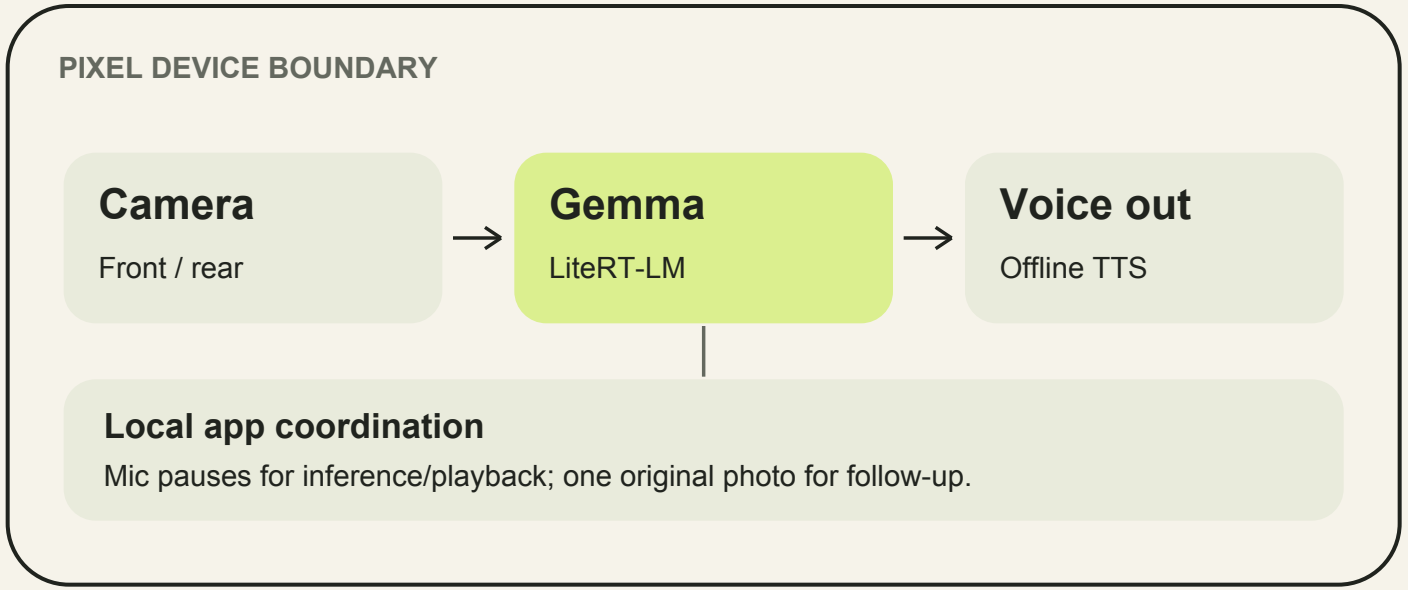
Prewarming moves work to startup/playback.
The smaller image budget trades detail for speed.
MTP was tested and left off after a runtime error.

Two runs per configuration on one saved photo; output length and device state vary. Timings exclude model load, microphone transcription, camera capture and speech. No guaranteed speedup.

BEST IN SHOW / END-TO-END ANDROID & PIXEL AI EXPERIENCE

A whole interaction, on one phone.

Native camera, foreground audio, local reasoning and spoken follow-up work together.



WHY IT MERITS CONSIDERATION

Native device integration

Both cameras, microphone lifecycle, captions and offline speech in a focused Android app.

Private inference by design

No Internet permission, no cloud model and no server dependency. GPU inference is implemented; TPU acceleration is future work.

VERIFIED TODAY

46 unit tests + build/lint; device camera, spoken output and same-photo follow-up checks.

Still to improve: live wake reliability, garment specificity and debug-media deletion controls.